Data Cleaning, Transformation and Modelling

Q1. What is Data Loading in Power BI and why is it considered the first step of analysis?

Ans– Definition:

Data Loading is the process of importing data from different sources into Power BI so that it can be used for analysis, reporting, and visualization.

What happens in Data Loading?

In Power BI, data can be loaded from multiple sources such as:

- Excel files
- SQL databases
- CSV files
- Web data

- Cloud services

This is usually done using Get Data option.

Why is Data Loading the First Step of Analysis?

1. Data is the foundation
   Without data, no analysis is possible. You must first bring data into Power BI.
2. Prepares data for further steps
   After loading, you can:
   - Clean data (remove errors, handle missing values)
   - Transform data (change format, split columns, etc.)
3. Enables visualization
   Charts, graphs, and dashboards can only be created after data is loaded.
4. Supports decision making
   Loaded data becomes the base for insights and business decisions.

Simple Flow of Analysis in Power BI:

1. Data Loading (First Step)

2. Data Cleaning & Transformation
3. Data Modeling
4. Data Visualization (Reports & Dashboards)

Easy Example:

Suppose you have a sales dataset in Excel.

- First, you load it into Power BI
- Then you clean and analyze it
- Finally, you create reports like sales charts

Without loading the data, none of these steps can happen.

Conclusion:

Data Loading is considered the first step of analysis because it brings raw data into Power BI, making it ready for processing, analysis, and visualization.

Q2.Explain the difference between "Load" and "Transform Data" in Power BI

Ans– 1. Load (Load Data)

Meaning:
"Load" means directly importing data into Power BI without making any changes.

Key Points:

- Data is brought as it is from the source
- No cleaning or modification
- Fast and simple process
- Used when data is already clean and ready

Example:
You have a clean Excel file → Click Load → Data appears in Power BI instantly

🔹 2. Transform Data

Meaning:
"Transform Data" means editing, cleaning, and shaping data before loading it into Power BI.

Key Points:

- Opens Power Query Editor
- You can:
  - Remove errors or duplicates
  - Split columns (like Name into First & Last)
  - Change data types

- ○ Filter rows
- Ensures data is accurate and analysis-ready

Example:
 Your dataset has missing values or wrong formats → Click Transform Data → Fix issues → Then load

Main Difference (Table Format)

Feature	Load	Transform Data
Purpose	Import data directly	Clean & prepare data
Changes Allowed	No	Yes
Speed	Fast	Takes time
Tool Used	Direct load	Power Query Editor
Best Use	Clean data	Raw/dirty data

Simple Understanding:

- Load = Direct use

- Transform Data = Clean first, then use

Conclusion:

“Load” is used when your data is already perfect, while “Transform Data” is used when your data needs cleaning and preparation before analysis.

Q3. What is a Fact Table and a Dimension Table? Give examples from the dataset.

Ans– 1. What is a Fact Table

A Fact Table is the main table in a dataset that contains quantitative (numeric) data used for analysis.
It stores measurable values like counts, totals, or amounts and is usually large.

In your dataset, Fact Table contains:

- Confirmed_Cases
- Recovered_Cases
- Deaths
- Active_Cases
- Tests_Conducted
- Hospitalized

Example from your data:

Country	State	Date	Confirmed_ Cases	Recovered_ Cases	Deaths
Brazil	State_A	43831	41240	56133	158

Here, all numeric columns (cases, deaths, etc.) are facts because they can be measured and analyzed.

2. What is a Dimension Table

A Dimension Table contains descriptive (text) information that helps to understand and categorize the data in the fact table.

In your dataset, Dimension Table contains:

- Country
- State
- Date
- Vaccination_Status

Example from your data:

Country	State	Date	Vaccination_Status
Brazil	State_A	43831	No

Q4. Why is Star Schema preferred over Snowflake Schema in Power BI?

Ans– In Power BI, the Star Schema is generally preferred because it is simpler, faster, and easier to use compared to the Snowflake Schema.

1. What is Star Schema vs Snowflake Schema

- Star Schema
  One central Fact Table connected directly to multiple Dimension Tables (no further splitting).
- Snowflake Schema
  Dimension tables are normalized (split into multiple related tables).

2. Reasons Why Star Schema is Preferred

1. Better Performance

- Star schema uses fewer joins

- Power BI can process queries faster using its engine (VertiPaq)

Snowflake schema requires multiple joins → slower performance

2. Simpler Data Model

- Star schema has a clean and simple structure
- Easy for beginners and analysts to understand relationships

Snowflake schema becomes complex due to multiple linked tables

3. Easy Report Creation

- Drag-and-drop becomes easier in Power BI
- Users don’t need to navigate multiple tables

Snowflake schema confuses users with too many tables

4. Better DAX Calculations

- Writing DAX formulas is easier with fewer relationships

- Less ambiguity in calculations

Snowflake schema increases chances of errors

5. Optimized for Power BI Engine

- Power BI is designed for denormalized models
- Star schema works best with in-memory compression

3. Example

Star Schema

```
         Country
            |
State — Fact Table — Date
            |
   Vaccination Status
```

Snowflake Schema

```
Country → Region → State → Fact Table
                 |
            Date Table
```

Q5. Identify and remove duplicate records based on Date, Country, and State.

Ans– What is a Duplicate Record

A duplicate record means same values appear more than once for:

- Country
- State
- Date

Example:
If two rows have:

- Country = India
- State = State_A
- Date = 43852

Then one of them is duplicate.

🔹 **Steps in Power BI**

Step 1: Load Data

- Open Power BI Desktop
- Click Home → Get Data → Excel/Text/CSV
- Load your dataset

Step 2: Open Power Query

- Click Transform Data
- Power Query Editor will open

Step 3: Select Columns

- Press Ctrl + Click and select:
    - Country
    - State
    - Date

These are the columns used to detect duplicates.

Step 4: Remove Duplicates

- Go to Home Tab
- Click Remove Rows → Remove Duplicates

✔ Power BI will:

- Keep the first occurrence
- Remove all duplicate rows based on selected columns

Step 5: Apply Changes

- Click Close & Apply

Final Table After Removing Duplicates

After removing duplicates based on Country, State, and Date, your table will look like this:

Country	State	Date	Confirmed	Recovered	Deaths	Active	Tests	Hospitalized	Vaccination
Brazil	State_A	43831	41240	56133	158	41424	123788	10288	No
UK	State_B	43832	83807	61268	1977	44002	98239	17709	No
Italy	State_B	43833	33434	38243	956	53744	27928	17543	Yes
UK	State_A	43834	69724	57384	2071	4795	281466	4650	(Blank)
USA	State_B	43836	28732	26657	4611	4798	351040	15200	Yes
India	State_A	43852	23624	11664	4850	26035	282590	16294	Yes

Q6.Identify and replace null values in Vaccination Status.

Ans– Solution in Power BI (Power Query Editor)

Step 1: Open Power Query

- Go to Home tab

- Click on Transform Data

Step 2: Select Column

- Click on Vaccination_Status column

Step 3: Identify Null Values

- Look for blank cells (they appear as empty or null)
- You can also:
  - Click the filter dropdown
  - Check `(null)` values

Step 4: Replace Null Values

Method 1 (Recommended):

- Right-click on the column
- Click Replace Values
- In dialog box:
  - Value to Find → `null`
  - Replace With → `"Unknown"` (or `"Not Vaccinated"`)

Method 2 (Using Transform):

- Go to Transform tab
- Click Replace Values
- Same steps as above

Step 5: Apply Changes

- Click Close & Apply

Final Table (Cleaned Data)

Country	State	Date	Confirmed_Cases	Recovered_Cases	Deaths	Active_Cases	Vaccination_Status
Brazil	State_A	43831	41240	56133	158	41424	No
UK	State_B	43832	83807	61268	1977	44002	No
Italy	State_B	43833	33434	38243	956	53744	Yes
UK	State_A	43834	69724	57384	2071	4795	Unknown
UK	State_C	43835	75697	31653	403	10634	Unknown
USA	State_B	43836	28732	26657	4611	4798	Yes
Italy	State_B	43837	11314	39952	2779	22420	No
Italy	State_B	43838	23954	22180	1387	51507	Unknown

Italy	State_A	43839	57108	78411	3453	55039	No
UK	State_D	43840	87044	1622	955	29300	Yes

Q7. Create a new column to calculate Recovery Rate

Ans– What is Recovery Rate?

Recovery Rate is calculated as:

Recovery Rate = Recovered Cases ÷ Confirmed Cases × 100

It shows the percentage of people who recovered out of total confirmed cases.

🔹 Steps to Create Recovery Rate Column in Power BI

Step 1:

Open Power BI and go to Data View

Step 2:

Click on New Column from the top ribbon

Step 3:

Write the following DAX formula:

Recovery Rate =
DIVIDE([Recovered_Cases], [Confirmed_Cases], 0) *
100

🔹 Why use `DIVIDE`?

- It prevents errors when Confirmed Cases = 0
- It is safer than normal division

🔹 **Final Table**

Country	**Confirmed_ Cases**	**Recovered_ Cases**	**Recovery Rate (%)**
Brazil	41240	56133	136.1%
UK	83807	61268	73.1%
Italy	33434	38243	114.4%
USA	28732	26657	92.7%

🔹 **Important Note**

- If Recovery Rate is greater than 100%, it means the data may be inconsistent
- In real scenarios, recovery rate is usually ≤ 100%

Q8. Create a summarized table showing total confirmed cases by Country

Ans– Method 1: Using Table Visual (Easy Way)

🔹 Steps:

1. Open Power BI Desktop
2. Go to Report View
3. Click on Table Visual
4. Drag:
   - Country → Values
   - Confirmed_Cases → Values
5. Click dropdown on Confirmed_Cases → Select Sum

Result:

- Power BI automatically groups by Country
- Shows total confirmed cases for each country

Method 2: Using DAX (Create New Table)

🔹 Steps:

1. Go to Modeling Tab

2. Click on New Table
3. Write this DAX formula:

Summary_Table =

SUMMARIZE(

YourTableName,

YourTableName[Country],

"Total Confirmed Cases", SUM(YourTableName[Confirmed_Cases])

Result:

- A new summarized table is created
- Shows total confirmed cases by country

Method 3: Using Measure (Best Practice)

🔹 Create Measure:

Total Confirmed Cases = SUM(YourTableName[Confirmed_Cases])

Then:

- Add Country in rows
- Add this measure in values

In Power BI, a summarized table showing total confirmed cases by country is created by adding a Table visual, placing the Country field, and applying the SUM aggregation on the Confirmed_Cases column. This groups the data by country and calculates the total confirmed cases for each country, providing a clear and concise summary for analysis.